

Jihoon Hong

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Atlanta, GA

RESEARCH INTERESTS

Uncertainty-aware computer vision techniques and control-theoretic tools for robust, reliable, and safe learning-enabled robotic systems, with applications in safe navigation and constrained generation.

EDUCATION

- **Georgia Institute of Technology** Aug. 2024 - Expected Aug. 2029
Ph.D., Machine Learning Atlanta, GA
- **Seoul National University** Mar. 2017 - Feb. 2024
B.A., Economics & B.S., Computer Science and Engineering Seoul, Korea
 - Summa Cum Laude (GPA: 3.93/4.0)
 - Minor in Mathematics

RESEARCH EXPERIENCE

- **Verifiable Optimization for Imaging and Learning Advances (VOILA) Lab** May 2025 - Present
Advised by Dr. Sara Fridovich-Keil Atlanta, GA
 - Developing uncertainty quantification and out-of-distribution detection methods using diffusion priors.
- **Trustworthy Robotics Lab** May 2025 - Present
Advised by Dr. Glen Chou Atlanta, GA
 - Improving robustness and performance of world/action models via control-theoretic analysis and intervention.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, U=UNDER REVIEW

* denotes equal contribution.

- [J.1] J. Hong, C. Chiu, S. Fridovich-Keil, G. Chou. (2026). **PolyMerge: Compressing 3D Gaussian Splats with Polytope Coverings for Provably Safe Resource-Constrained Navigation**. *IEEE Robotics and Automation Letters (RA-L)*. DOI: 10.1109/LRA.2026.3692083
- [C.3] A. Kheirandish*, J. Hong*, S. Fridovich-Keil. (2026). **KLIP: localized distribution shift detection via KL-divergence with diffusion priors in Inverse Problems**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [C.2] S. Lee, H. Lee, J. Hong, S. Cho, J. Lee. (2024). **VGA: Hardware Accelerator for Scalable Long Sequence Model**. *57th International Symposium on Microarchitecture (MICRO)*.
- [C.1] H. Lee, J. Hong, S. Kim, S. Lee, J. Lee. (2023). **A Memory-Efficient Edge Inference Accelerator with XOR-based Model Compression**. *60th Design Automation Conference (DAC)*.
- [U.3] J. Hong, J. Skifstad, Q. Dai, A. Chan, G. Chou. (2026). **Steering Robustness into World Action Models via Mechanistic Interpretability and Optimal Control**. Manuscript submitted to the *Conference on Robot Learning (CoRL)*.
- [U.2] J. Hong, A. Chan, Q. Dai, J. Skifstad, G. Chou. (2026). **Activation Steering of Video Generation Models via Reduced-Order Linear Optimal Control**. Manuscript submitted to the *40th Annual Conference on Neural Information Processing Systems (NeurIPS)*.
- [U.1] J. Hong*, A. Kheirandish*, S. Fridovich-Keil. (2026). **LACToS: Local Anomaly detection via Conditional Typicality of Superpixels**. Manuscript submitted to the *40th Annual Conference on Neural Information Processing Systems (NeurIPS)*.

HONORS AND AWARDS

- **Kwanjeong Educational Foundation (KEF) Scholarship** Sep. 2024 – Aug. 2029